

HAT-P-36b

R-0981

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_220320 · created 2026-07-29T20:32:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459658.7266 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.046
Transit depth (Rp/Rs)^2	2.44 +/- 1.44 [%]
Orbital Inclination (inc)	89.41 +/- 2.71
Airmass coefficient 1 (a1)	1.684 +/- 0.073
Airmass coefficient 2 (a2)	-0.49 +/- 0.12
Scatter in the residuals of the lightcurve fit is	4.3 %
Best Comparison Star	#1 - [289, 117]
Optimal Aperture	2.37
Optimal Annulus	11.46
Transit Duration (day)	0.052 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.156 $\pm$ 0.046 vs 0.126 (deviation 0.42 σ)
Duration fit vs published	0.052 d vs 0.0925 d (deviation 5.26 σ)
Mid-time O-C	10.2 min
Depth SNR	2.2
Residual scatter	4.3 %

Light curve

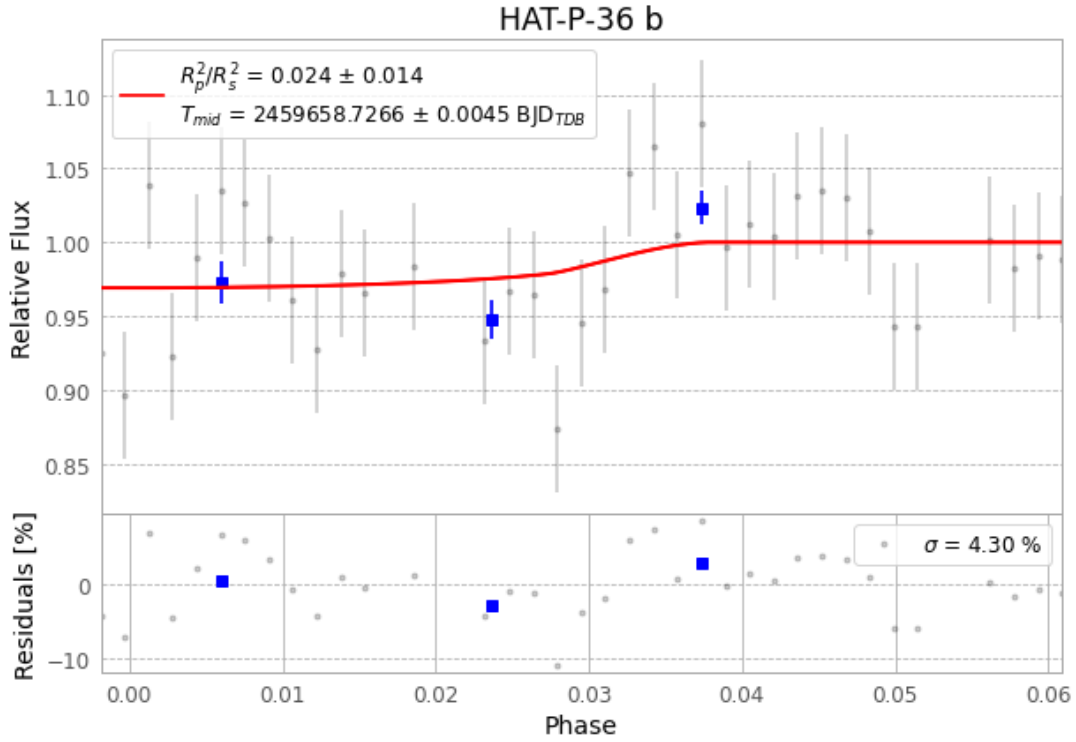


Figure 1 — FinalLightCurve_HAT-P-36 b_2022-03-19.png (SHA-256 9ff03ce431ea949f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [180, 244], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aa42a6ca5d9d645c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

37 FITS frames (24 MB), HATP-36220320051515.FITS → HATP-36220320071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-220320.log (5fd8a749c4d9...), FinalLightCurve_HAT-P-36 b_2022-03-19.png (9ff03ce431ea...), FinalLightCurve_HAT-P-36 b_2022-03-19.csv (4fa441480026...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 20:32Z.